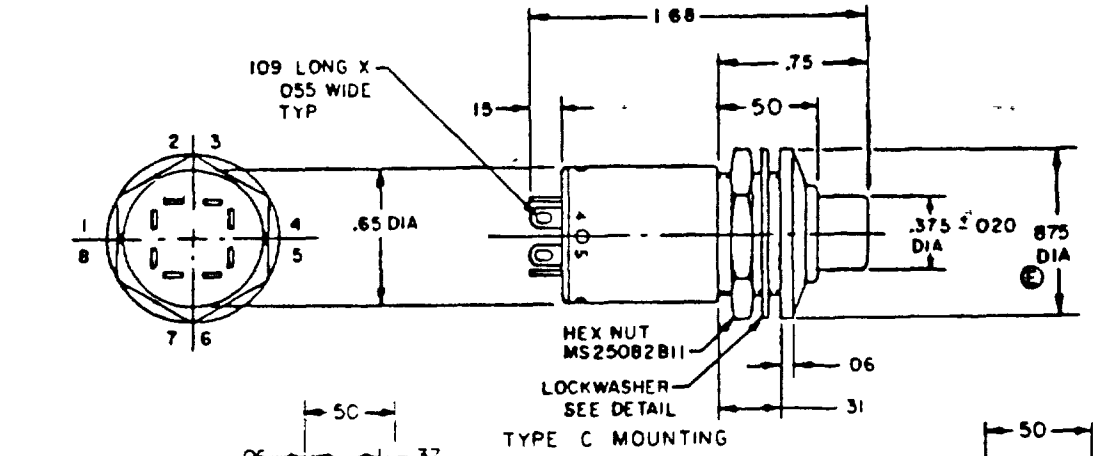
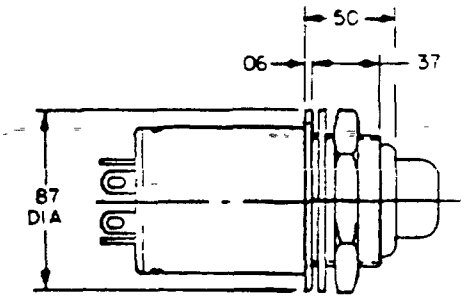
MILITARY SPECIFICATION SHEET

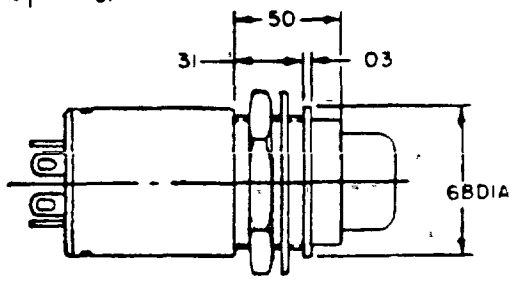
Ⓔ SWITCHES, PUSH, MOMENTARY, 4 CIRCUIT, 4 AMPERES OR LOW LEVEL, DUSTTIGHT

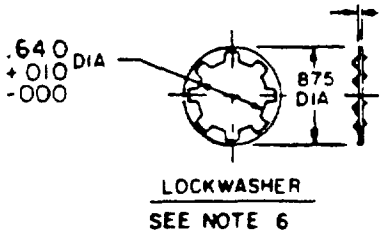
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-8805.

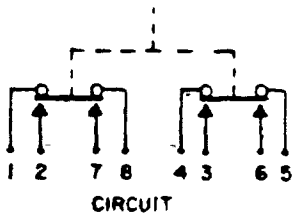








Inches	mm	Inches	mm
.010	0.25	.37	9.4
.020	0.51	.375	9.53
.022	0.56	.50	12.7
.03	0.8	.640	16.26
.055	1.40	.65	16.5
.06	1.5	.68	17.2
.109	2.77	.75	19.1
.15	3.8	.87	22.1
.25	6.4	.875	22.23
.31	7.8	1.68	42.7



NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.02 (0.51 mm) for two place decimals and ±.005 (0.13 mm) for three place decimals.
3. The circuit and terminal arrangement shall be permanently and legibly marked on the switch case.
4. Metric equivalents are given for general information only.
5. Mounting hardware has .625-24 UNEF-2A external thread and is removable for installation.
Ⓔ 6. Lockwasher may be replaced by MS35333-138.

FIGURE 1. Dimensions and configurations

Ⓔ denotes changes

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DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Enclosure design: 2 (dusttight).

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: Method 213, test condition A, MIL-STD-202 (50 G sine wave).

Ⓔ Sinusoidal vibration grade: 1 (10 to 500 Hz).

Weight: .040 pound maximum.

Operating characteristics:

Total plunger travel: .085 ±.015 inch.

Operating force: 4 ±1 pounds.

Strength of actuator means: 25 pounds.

Terminal strength 5 pounds.

Material:

Ⓔ Plunger: May be glass filled nylon.

Case: Aluminum alloy.

Lockwasher Phosphor bronze, black oxide finish.

Back Flame resistant diallyl phthalate which meets the requirements of MIL-M-14.

Finish: Case, anodized in accordance with MIL-A-8625, type II. Exposed surfaces, as viewed by the operator, shall be black anodized

Dielectric withstanding voltage:

Sea level: 1,000 V rms. In qualification inspection table after electrical endurance the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable.

Mechanical endurance: 50,000 cycles.

Electrical endurance 25,000 cycles.

Electrical ratings: See table I.

Sand and dust: Applicable.

TABLE I. Electrical ratings, power circuit switches (MS27903-1 through -6).

Load	Sea level	
	28 V dc	115 V ac, 400 Hz
	amperes	amperes
Resistive - - - -	4	4
Inductive - - - -	2	2
Motor - - - - - -	2	2

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Ⓔ Low level circuit: Applicable when specified (see table III).

Ⓔ Electronic logic circuit (5 0 V dc, .010 ampere): Applicable when specified (see table III): 25,000 cycles at an actuation rate of 120 cycles per minute maximum with no "stick", or "misses", allowed when tested in accordance with ANSI/EIA RS448, method 17 as follows:

Test condition:

Each switch contact shall be tested using a 5.0 ±0.5 V dc, 10 ±1 mA resistive load. During each closure of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact static closure. The switch contacts need not be monitored until 10 milliseconds after the initial contact closure to exclude any contact bounce. During each opening of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact opening.

A voltage of 2.1 volts or greater across the switch terminals shall constitute a contact "miss" (failure to properly close the circuit). A voltage drop of less than 90 percent of the open-circuit voltage shall constitute a contact "stick" (failure to properly open the circuit).

The monitoring device shall either record the number of contact closures at which "sticks" and/or "misses" occur, or discontinue the test when "sticks" and/or "misses" occur.

Operating temperature:

a. 25 percent of the test cycle at -55°C specified.

b. 25 percent of the test cycle at room ambient conditions.

c. 50 percent of the test cycle at +85°C.

Qualification:

Group submission: See table II.

Ⓔ TABLE II. Qualification inspection.

Examination or test	Basic switch sample	Other switch samples	Extent of approval	
Qualification inspection table of MIL-S-8805	MS27903-1 or MS27903-2 (all sample units)	MS27903-3, -4, -5, and -6 (2 sample units each). Visual and mechanical examination	MS27903-1 through 06	
Group I and group X of the qualification inspection table of MIL-S-8805 plus 2 additional samples shall be subjected to logic level circuit and operating characteristics	MS27903-7 or -8			All

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Ⓔ Part number: See table III.

TABLE III. Part numbers and characteristics.

Part number	Type	Button	Electrical ratings
MS27903-1	C	Black	Power circuit rating (see table I)
MS27903-2	C	Red	
MS27903-3	G	Black	
MS27903-4	G	Red	
MS27903-5	F	Black	
MS27903-6	F	Red	
MS27903-7	C	Black	Low level and logic level rating
MS27903-8	C	Red	
MS27903-9	G	Black	
MS27903-10	G	Red	
MS27903-11	F	Black	
MS27903-12	F	Red	

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities.
Army - AR
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities.
Army - AV, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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